Designing FDCAN bus

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$t_{sync} = 1 \times t_q$ (Sync)

Propagation segment - to compensate for delays induced by signal propagation

Phase segment 1 & 2

- to compensate for changes of edge position of signals due to frequency of various nodes.

Bit segment 1 (BS1)

$t_{BS1} = t_{prop-seg} + t_{ph-seg1}$

$t_{BS2} = t_{ph-seg2}$ ⟶ defines, with t_{BS1} the sample point position.

$$\text{Can bit rate} = \frac{1}{\text{Nominal CAN bit time}}$$

If time quanta = 11, $t_q = 0.909\,\mu s$ so

CAN bit time $= 0.0909\,\mu s \times 11 = 1\,\mu s$.

CAN bit rate $= \frac{1}{1\,\mu s} = 1$ Mbps.

Time Quanta – unit of bit time.

$$T_q\,(\text{second}) = \frac{\text{Prescaler}}{\text{CAN frequency}}$$

If Nominal prescaler = 1. CAN clock = 120 MHz

Required : Sample point of 0.66

$$T_q\,(ns) = \frac{1}{120 \times 10^6} = 8.33$$

How long is 1 bit of time?

$$1\text{ Mbps} = \frac{1}{\text{CAN bit of time}}$$

CAN bit of time = 1000 ns

How many TQ in 1 bit?

$$TQ = \frac{1000}{8.33} = 120 \text{ Tq per bit}$$

Sample point at approx 66.6 %.

$120 \times 0.666 = 79.92 \approx 80$

Sync seg = 1 (fixed)
Nom Time seg 1 = 79
} $\frac{1+79}{120} \times 100 = 66.7\,\%$

Nom Time seg 2 = $(120 - 1 - 79) = 40$

For Data rate (5 Mbps)

CAN bit time $= \frac{1}{5 \times 10^6} = 200$ ns

Total Tq $= \frac{200}{8.33} = 24$ Tq per bit

Sample point at approx 66.6 %

$24 \times 0.666 = 15.984 \approx 16$

Sync seg = 1 (fixed)
Data time seg 1 = 15
} $\frac{1+15}{24} \times 100 = 66.7\,\%$

Data time seg 2 = $24 - 1 - 15 = 8$

Conclusion :

120 MHz CAN clock w/ BRS (1/5 Mbps)

Nom phase (66.7 % sample point)

Parameter	value
prescaler	1
TS1	79
TS2	40
SJW	40

Data phase (66.7 % sample point)

Parameter	Value
prescaler	1
TS1	15
TS2	8
SJW	8

* Sync Jump width : max amt of time in Tq, the controller can shorten or lengthen a single bit to stay synchronised w/ other nodes.

$$1 \leq SJW \leq \min(TSEG2, Register_limit)$$

If SJW = 1, controller can only adjust bit timing by 1 Tq, so if signal drifts by 2Tq, the controller loses sync.
If SJW = 16, controller can 'stretch' or 'shrink' the bit to catch a drifting edge.